Stephen Ni

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Education

University of Waterloo Sept. 2023 – April. 2027
Systems Design Engineering (GPA: 3.92/4.0) *Waterloo, ON*

Technical Skills

Languages: Python, Java, Ruby, TypeScript, SQL, C++, Go, Bash, Haskell
Frameworks: Next.js, Vue.js, Express.js, FastAPI, Flask, Django, Rails, LangChain, Spring Boot
Developer Tools: AWS, GCP, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, JFrog, Redis, Snowflake, BigQuery, Kafka, RabbitMQ, Postman, Nginx, Splunk, Firebase, Supabase, Jira

Experience

Shopify May 2025 – August 2025
Software Engineering Intern *Toronto, Ontario*

- Engineered a full-stack USDC payment gateway using **Ruby on Rails** and **React**, orchestrating transaction flows between Shopify, Coinbase, and Stripe to introduce a new payment method.
- Eliminated critical race conditions between on-chain and off-chain states by implementing database locks, idempotency keys, and ERC3009 escrow contracts, ensuring over **99.9%** payment accuracy.
- Built an automated gating system for the iDEAL payment method using scheduled jobs and event-driven actions to manage merchant access based on real-time risk criteria, reducing fraudulent chargeback rates to below **0.5%**.

Trend Micro September 2024 – December 2024
Software Engineering Intern *Ottawa, Ontario*

- Developed a retrieval-augmented generation (RAG) chatbot using **LangChain** and **ChromaDB**, parsing internal documentation and improving ROUGE/BLEU scores by **25%** for compliance related queries.
- Built a resilient data pipeline with **Spring Boot** and **AWS SQS** to process **MySQL** change events, leveraging a **Dead-Letter Queue (DLQ)** and automated retries to ensure data integrity.
- Created a multi-stage CI/CD pipelines with **Jenkins**, **Terraform**, and **JFrog**, improving stage-specific testing

Dropbase January 2024 – April 2024
Software Engineering Intern *San Francisco, California*

- Revamped the internal tool builder's frontend using **React**, **Jotai**, and **Chakra UI**, creating a more responsive and maintainable user interface with efficient data fetching via **Axios**.
- Created a database abstraction layer to extend backend support from **one** to **four** major DBMSs (Postgres, MySQL, Snowflake, SQLite), significantly broadening product compatibility.
- Architected a database connection pooling layer using **PgBouncer** hosted on **AWS EC2**, boosting SQL query performance by **60%**.

Projects

Screentime | *SwiftUI, Supabase, PostgreSQL* June 2025 – August 2025

- Built an iOS screen time app with **SwiftUI** and the Family Controls API, structuring its architecture around the **MVVM** design pattern and **SOLID** principles to ensure scalability.
- Implemented an asynchronous job queue for offline functionality, allowing chore completions and time usage to be cached on-device and synced upon reconnection.
- Architected a secure backend in **PostgreSQL**, integrating atomic transactions and row-level security to ensure the integrity and isolation of user data.

ConnectX | *FastAPI, React, PostgreSQL, Docker* July 2023 – September 2023

- Developed a full-stack web application with a **FastAPI** backend and **React** frontend, implementing secure user authentication and data persistence via **JWT** tokens and **PostgreSQL**.
- Established an end-to-c CI/CD pipeline using **GitHub Actions** and **Docker**, automating the testing and containerized deployment of the application to a Digital Ocean server configured with **Nginx**.

Celeste | *Go* July 2023 – August 2023

- Developed a Turing-complete compiler in **Go**, comprising of a **lexer**, **parser**, and **abstract syntax tree**
- Created a **compiler** designed to compile nodes in an abstract syntax tree into byte code
- Implemented a **virtual machine** capable of reading and executing the generated byte code